



Microprocessors And Microcontrollers

Assignment 1- Week 1

QUESTION 1:

Correct Answer: b

Detailed Solution:

Step 1: Convert the numbers to binary

1. **2.625** in binary:
 - $2 = 10210_2102$
 - $0.625 = 0.10120.101_20.1012$
 - So, $2.625 = 10.101210.101_210.1012$
2. **10.25** in binary:
 - $10 = 101021010_210102$
 - $0.25 = 0.0120.01_20.012$
 - So, $10.25 = 1010.0121010.01_21010.012$

Step 2: Convert to 2's complement

To subtract $10.2510.2510.25$ from $2.6252.6252.625$, we need to represent $-10.25-10.25-10.25$ in 2's complement form.

1. **Represent 10.25:**
 - $10.25=1010.01210.25 = 1010.01_210.25=1010.012$
2. **Invert the bits** for $1010.0121010.01_21010.012$:
 - $1010.012 \rightarrow 0101.1021010.01_2 \rightarrow 0101.10_21010.012 \rightarrow 0101.102$
3. **Add 1** to the result:
 - $0101.102+1=0101.1120101.10_2 + 1 = 0101.11_20101.102+1=0101.112$

So, the 2's complement representation of $-10.25-10.25-10.25$ is $0101.1120101.11_20101.112$.

Step 3: Perform the subtraction using 2's complement

Now, to subtract $10.2510.2510.25$, we add $2.6252.6252.625$ to the 2's complement of $10.2510.2510.25$:

- $2.625=10.10122.625 = 10.101_22.625=10.1012$
- $-10.25=0101.112-10.25 = 0101.11_2-10.25=0101.112$



Add these two numbers:

$10.1012 + 0101.112$ $10.101_2 + 0101.11_2$ $10.1012 + 0101.112$

Perform the binary addition:

- Add 101101101 and 110110110 : this results in 101121011_2 , and carry over the 1.

The result will be $10000.011210000.011_2$, which is the binary representation of the result.

Thus, the correct answer is **(11000.011)₂**, which corresponds to option **b**.

QUESTION 2:

Correct Answer: a

Detailed Solution:



Step 1: Convert +1 to binary

In 4-bit binary, +1 is represented as:

$$0001_2$$

Step 2: Find the 2's complement to represent -1

To get the 2's complement (which represents negative numbers), perform the following:

1. Invert the bits of 0001_2 , which gives:

$$1110_2$$

2. Add 1 to the result:

$$1110_2 + 1 = 1111_2$$

So, -1 in 4-bit 2's complement is 1111_2 .

QUESTION 3:

Correct Answer: a

Detailed Solution:

The range of values for n-bit number in 2's complement system is given as -2^{n-1} , $(2^{n-1} - 1)$. Substitute $n = 4$, we get the range as -8 to +7.

QUESTION 4:

Correct Answer: c

Detailed Solution:

The decimal equivalent would be $(1 \times 2^0) + (1 \times 0.5^1) + (1 \times 0.5^2) = \frac{3}{4}$



QUESTION 5:

Correct Answer: c

Detailed Solution:

QUESTION 6:

Correct Answer: b

Detailed Solution:

QUESTION 7:

Correct Answer: c

Detailed Solution:

QUESTION 8:

Correct Answer: d

Detailed Solution:

QUESTION 9:

Correct Answer: d

Detailed Solution:



4K RAM need 12 address lines to access particular location, and CPU address width is 16 bit. Address lines A11 to A0 of CPU is connected to address bus of the memory, out of remaining 4 address lines of CPU 3 lines (A15'A14A13) connected to chip select(CS) of the RAM. Address line A12 can be 0 or 1, so the possible address range:

for A12 = 0 : 6000-6FFFH and

for A12 = 1 : 7000-7FFFH

QUESTION 10:

Correct Answer: a

Detailed Solution:

*****END*****